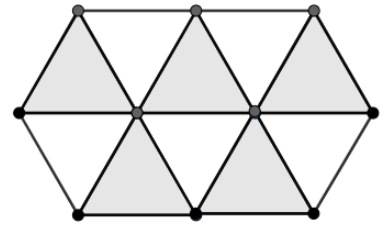
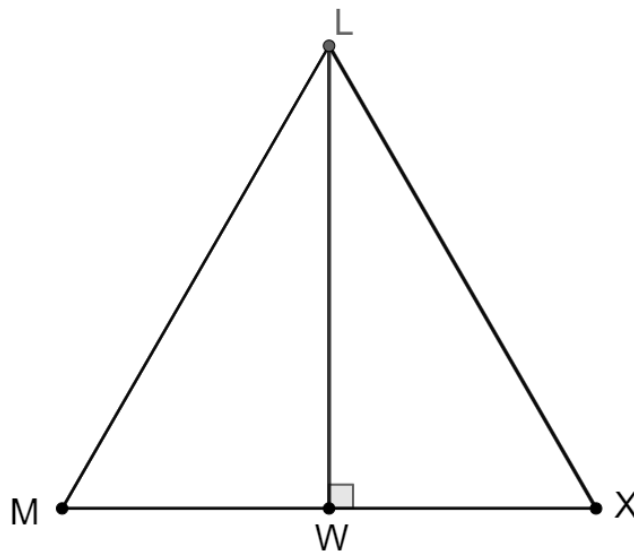


1. An architect is designing a building that will use a series of equilateral triangles to create a unique window. A sketch of the design is shown.



Equilateral triangle LXM models one of the equilateral triangles in the window. The height of the triangle, $\overline{LW}$, is 2 feet (ft).

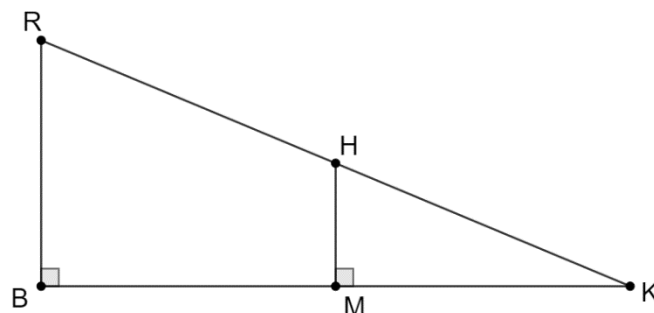


Determine the perimeter, in feet, of one of the triangles in the window. Round your answer to the nearest thousandth.

Work	
Perimeter	6.928 feet

2. Right triangles HMK and RBK are shown.

The measurements of the line segments in the diagram are shown in the table.



Line Segment	Measurement	Line Segment	Measurement
MH	8	MB	19.2
MK	19.2	RH	20.8
KH	20.8	RB	16

- a. Using angle K as the reference angle, what is the ratio $\frac{\text{adjacent}}{\text{hypotenuse}}$ for $\triangle RBK$.

$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{38.4}{41.6}$
---	---------------------

- b. Using angle K as the reference angle, what is the ratio $\frac{\text{opposite}}{\text{adjacent}}$ for $\triangle HMK$.

$\frac{\text{opposite}}{\text{adjacent}}$	$\frac{8}{19.2}$
---	------------------

- c. Complete the statements.

By using angle K as the reference angle, it can be shown that the ratio

$\frac{\text{opposite}}{\text{hypotenuse}}$ for $\triangle HMK$

- ☒ is
☐ is not

equivalent to the ratio $\frac{\text{opposite}}{\text{hypotenuse}}$ for $\triangle RBK$.

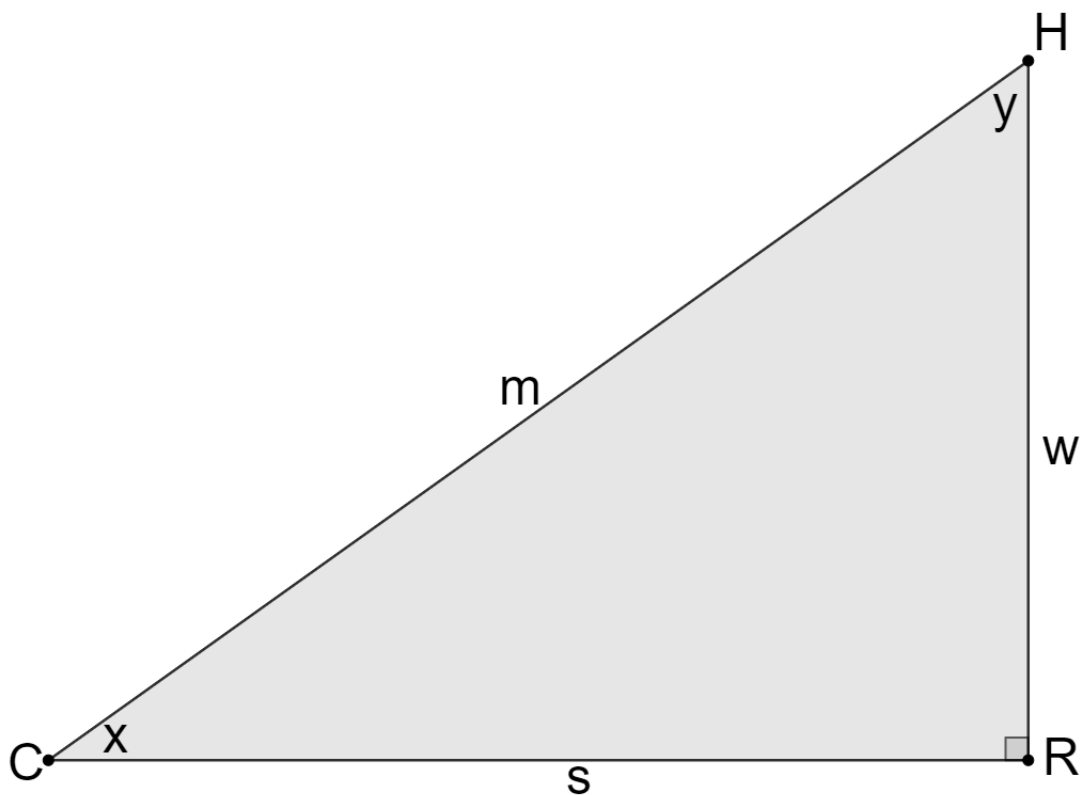
- ☒ All
☐ No

similar triangles that use angle K as the reference angle

- ☒ will
☐ will not

have an equivalent $\frac{\text{opposite}}{\text{hypotenuse}}$ ratio.

3. $\triangle HRC$ is shown where $s = 9$, $m = \sqrt{125}$, $x = 36.4^\circ$.



Select all of the equations that can be used to determine the value of w .

☐ $\tan(x) = \frac{9}{w}$

☐ $9^2 + w^2 = (\sqrt{125})^2$

☐ $\cos(y) = \frac{w}{\sqrt{125}}$

☐ $w = \sqrt{9^2 + (\sqrt{125})^2}$

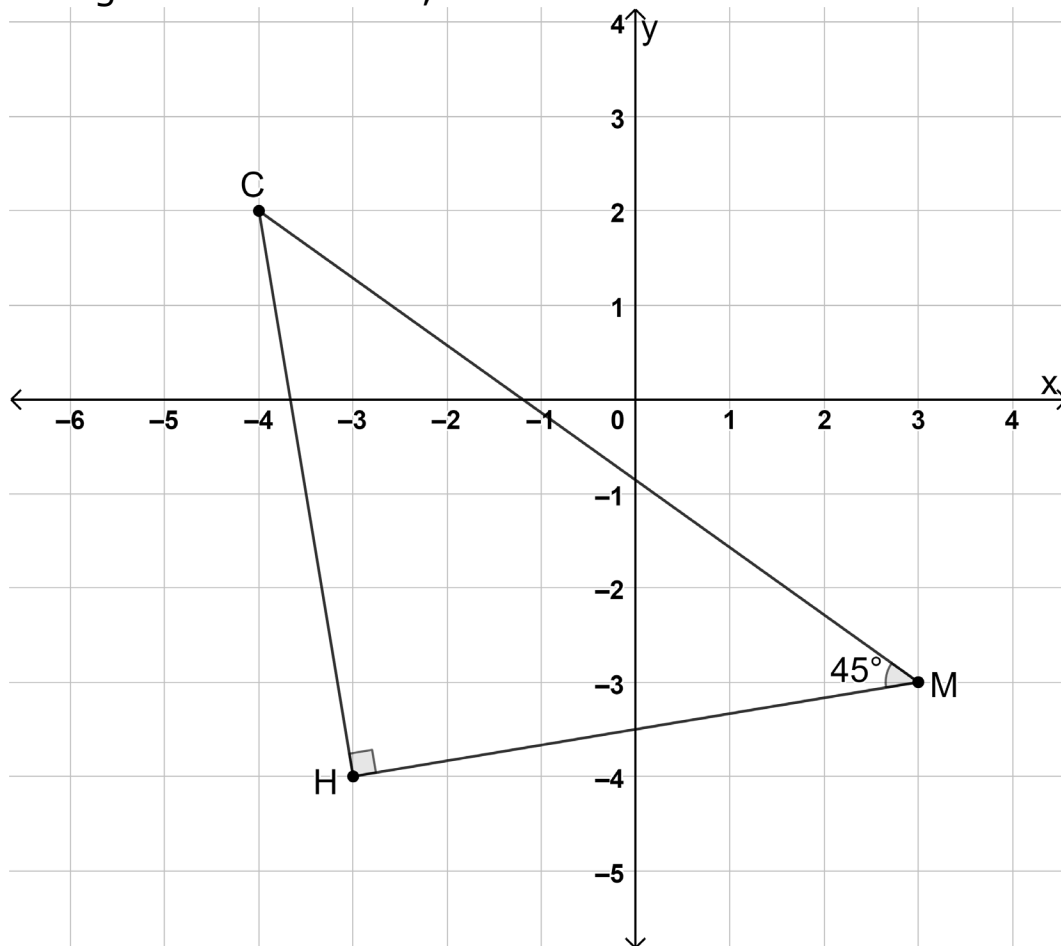
☐ $\sin(90 - x) = \frac{9}{w}$

☐ $\sin(36.4^\circ) = \frac{w}{\sqrt{125}}$

☐ $\cos(53.6^\circ) = \frac{w}{\sqrt{125}}$

☐ $\tan(90 - x) = \frac{9}{w}$

4. Right triangle MHC is shown, where $m\angle CMH = 45^\circ$.



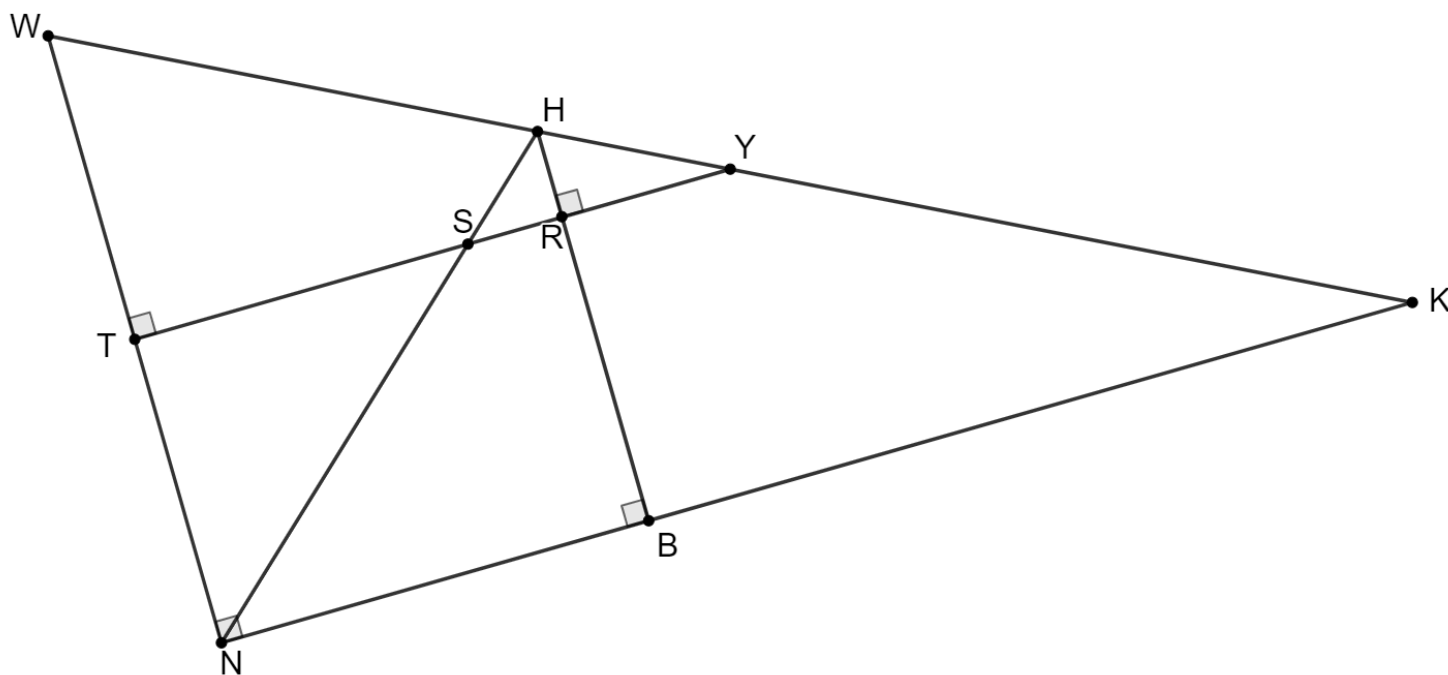
- a. Determine the length of $\overline{CH}$. Leave your answer in exact form.

Work	
CH =	$\sqrt{37}$

- b. Determine the length of $\overline{CM}$. Leave your answer in exact form.

Work	
CM =	$\sqrt{74}$

5. Triangle WKN is shown. Some of the measurements are given.



$$m\angle WTY = 90^\circ$$

$$m\angle TWH = 63^\circ$$

$$m\angle HRY = 90^\circ$$

$$m\angle TSN = 42^\circ$$

$$m\angle HBN = 90^\circ$$

$$HY = 3.2$$

$$m\angle WNB = 90^\circ$$

$$TN = RB = 5.2$$

$$YK = 11.4$$

$$WH = 8.2$$

- a. Determine the length of $\overline{WN}$. Show your work. Round your answer to the nearest tenth.

Work	
WN =	10.4

- b. Determine the length of $\overline{HR}$. Show your work. Round your answer to the nearest tenth.

Work	
HR =	1.5

- c. Determine the length of $\overline{TS}$. Show your work. Round your answer to the nearest tenth.

Work	
TS =	5.8

6. Some of the longest zip lines in the US are in Marion County. Bronson chooses to ride the zip line that is 1,150 feet long. The angle of depression of the zip line is 10° . What is the height, in feet, of the start of the zip line? Show your work. Round your answer to the nearest thousandth.

Work	
Height	199.695 feet